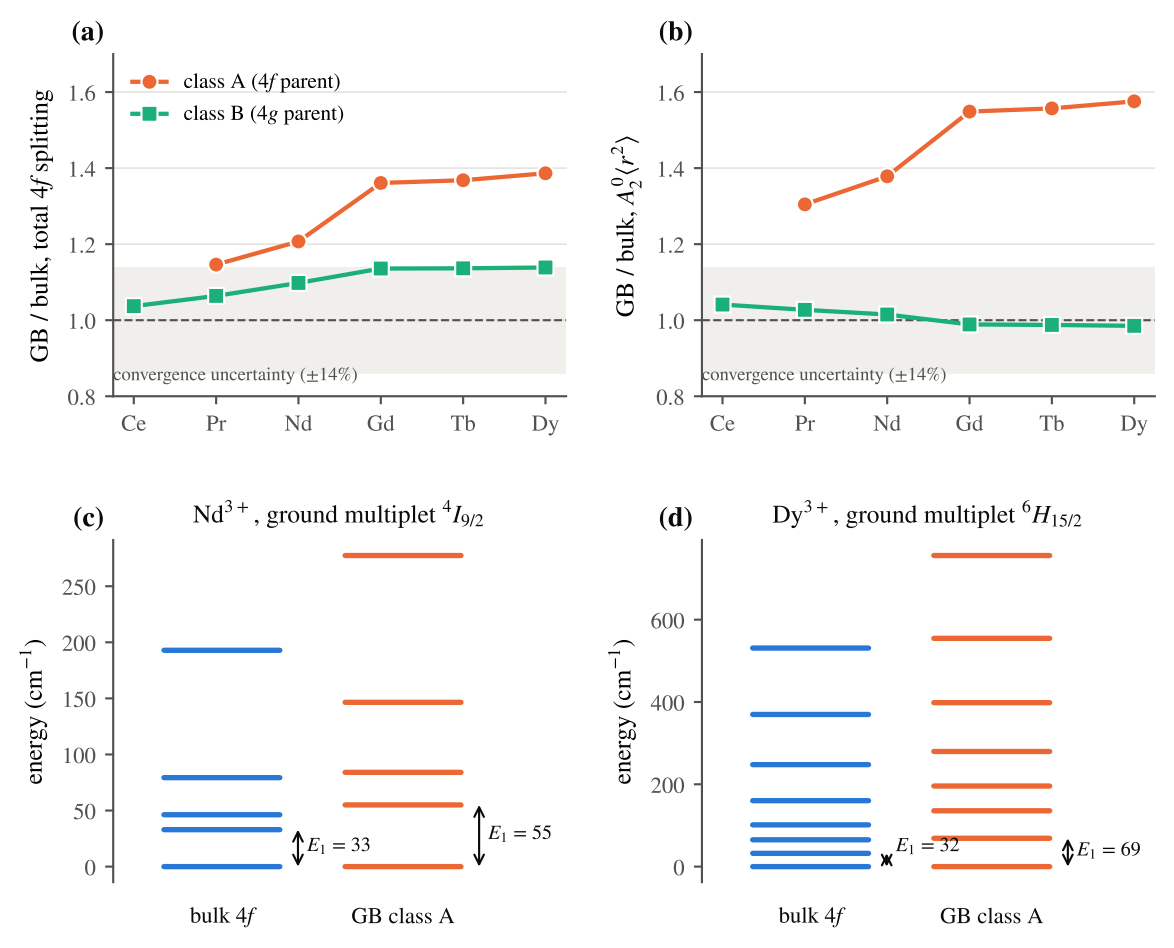
(a)
class A (4f parent)
class B (4g parent)
GB / bulk, total 4f splitting
1.6
1.4
1.2
1.0
0.8
convergence uncertainty (±14%)
Ce
Pr
Nd
Gd
Tb
Dy
(b)
GB / bulk, $A_2^0\langle r^2\rangle$
1.6
1.4
1.2
1.0
0.8
convergence uncertainty (±14%)
Ce
Pr
Nd
Gd
Tb
Dy
(c)
Nd^{3+}, ground multiplet $^4I_{9/2}$
energy (cm^{-1})
250
200
150
100
50
0
$E_1 = 33$
$E_1 = 55$
bulk 4f
GB class A
(d)
Dy^{3+}, ground multiplet $^6H_{15/2}$
energy (cm^{-1})
600
400
200
0
$E_1 = 32$
$E_1 = 69$
bulk 4f
GB class A